Lab Test: Smart Washing Machine Microcontroller Design

(FSM + Interrupts + Cycle-Based Simulation)

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1. Objective

This test evaluates your ability to:

- Design a microcontroller architecture
- Model control logic using FSM
- Handle interrupts
- Perform cycle-by-cycle simulation

2. System Description

You are designing a controller for a **Smart Washing Machine**.

Inputs

- START
- DOOR_CLOSED (1 = closed, 0 = open)
- WATER_LEVEL_OK (1 = sufficient water)
- OVERLOAD (1 = overload condition)

Outputs

- MOTOR (ON/OFF)
- VALVE (ON/OFF)
- DOOR_LOCK (ON/OFF)
- BUZZER (ON/OFF)

3. Operating Modes

Mode	Wash Duration (cycles)
QUICK	2
NORMAL	4
HEAVY	6

4. FSM States and Meaning

State	Description
IDLE	Waiting for START
LOCK	Lock door (2 cycles)
FILL	Fill water until WATER_LEVEL_OK=1
WASH	Motor ON for required cycles
DRAIN	Stop water/motor
UNLOCK	Unlock door (2 cycles)
BUZZ	Indicate completion
ERROR	Safety stop state

5. State Output Behavior

State	MOTOR	VALVE	DOOR_LOCK	BUZZER
LOCK	OFF	OFF	ON	OFF
FILL	OFF	ON	ON	OFF
WASH	ON	OFF	ON	OFF
DRAIN	OFF	OFF	ON	OFF
UNLOCK	OFF	OFF	OFF	OFF
BUZZ	OFF	OFF	OFF	ON
ERROR	OFF	OFF	ON	ON

6. Timing Rules (Very Important)

- Each state takes 1 cycle unless specified
- LOCK and UNLOCK take 2 cycles
- WASH duration depends on mode
- State changes occur at the **end of each cycle**

7. Interrupt Handling

Interrupt priority:

`OVERLOAD > DOOR_OPEN > NORMAL TRANSITION`

Behavior

- OVERLOAD = 1 → go to ERROR immediately
- DOOR_CLOSED = 0 → go to ERROR immediately
- Interrupt overrides all states

8. Input Behavior

Inputs may change during execution.
Example:

```
Cycle 3: WATER_LEVEL_OK becomes 1
Cycle 5: OVERLOAD becomes 1
```

You must read inputs at every cycle.

9. Execution Model (Step-by-Step)

At each cycle:

1. Read inputs
2. Check interrupts
3. Execute current state
4. Update outputs
5. Transition to next state

10. Output Format (Mandatory)

```
Cycle | State | Inputs | Outputs
```

Example:

```
Cycle 3 | FILL | W=0 | VALVE=ON
```

11. Worked Example (Important)

```
MODE = QUICK
Cycle 1: START=1, DOOR=1, WATER=1
```

Execution

```
Cycle 1 | LOCK | DOOR_LOCK=ON
Cycle 2 | LOCK | DOOR_LOCK=ON
Cycle 3 | FILL | VALVE=ON
Cycle 4 | WASH | MOTOR=ON
Cycle 5 | WASH | MOTOR=ON
Cycle 6 | DRAIN | MOTOR=OFF
```

12. Sample Test Cases

Test Case 1 (Normal)

```
MODE = NORMAL
Cycle 1: START=1, DOOR=1, WATER=1
```

Expected Key Output

```
Cycle 4-7 | WASH | MOTOR=ON
Cycle 8 | DRAIN
```

Test Case 2 (Water Delay)

```
Cycle 1: WATER=0
Cycle 3: WATER=1
```

Expected

```
FILL continues until WATER=1
Then transition to WASH
```

Test Case 3 (Door Interrupt)

```
Cycle 4: DOOR=0
```

Expected

```
Immediate transition to ERROR
MOTOR=OFF, BUZZER=ON
```

Test Case 4 (Overload)

```
Cycle 6: OVERLOAD=1
```

Expected

```
Immediate ERROR state
```

13. Tasks

- Draw architecture diagram
- Draw FSM diagram
- Write C/C++ simulator
- Print cycle-by-cycle output

14. Evaluation

Component	Marks
Architecture	20
FSM	20
Interrupt Logic	20
Simulation Code	20
Output	10
Viva	10
Total	100

15. Important Assumptions

- All signals are synchronous per cycle
- Only one state is active at a time
- Outputs depend only on current state